IMAGE CLASSIFICATION AND EXPLAINABLE IDENTIFICATION OF AI-GENERATED SYNTHETIC IMAGES

Dual-Model Deep Learning with Grad-CAM Explainability

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Focus Group: Image Authentication & Explainable AI Team

CHAPTER 1: INTRODUCTION

Project Background

With the rapid proliferation of highly capable generative artificial intelligence models, the digital ecosystem is experiencing an influx of hyper-realistic synthetic media. Distinguishing between authentic, camera-captured photographs and synthetically generated images has become increasingly difficult for human observers. This phenomenon raises critical concerns regarding digital forgery, misinformation, and intellectual property infringement.

Problem Statement

There is an urgent requirement for reliable, automated computational tools to verify the authenticity of digital images. Current manual verification methods are subjective and inefficient. Furthermore, many existing deep learning solutions operate as opaque "black boxes," providing predictions without transparent contextual evidence, which diminishes user trust in automated systems.

Objectives

1. To develop a robust deep learning classifier capable of distinguishing between real and artificial intelligence-generated images with high accuracy.

2. To impleme
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Scope of the Project

The project encompasses the training, evaluation, and deployment of a MobileNetV2-based convolutional neural network on the CIFAKE dataset. The scope includes the development of a Streamlit frontend application and the integration of Gradient-weighted Class Activation Mapping (Grad-CAM) for explainability. The application is limited to binary classification (Real vs. AI Generated) on standard two-dimensional image formats.

Expected Outcomes

The expected outcome is a fully functional web-based application that accepts image uploads, accurately predicts authenticity, displays statistical confidence metrics, and provides a Grad-CAM heatmap visualization explaining the localization of discriminative features.

CHAPTER 2: SYSTEM OVERVIEW

Existing System

Existing systems for synthetic image detection frequently rely on metadata analysis, which is highly susceptible to manipulation. Alternative commercial deep learning models output simple probability scores without explicit reasoning, preventing users from understanding the specific features that triggered a positive or negative detection. Furthermore, most systems use a single generic model that performs poorly on specialized domains such as photorealistic deepfake faces.

Proposed System

The proposed system bridges these gaps by deploying a dual-model architecture: a lightweight MobileNetV2 model optimized for general objects (trained on CIFAKE), and a high-capacity EfficientNetV2B0 model optimized specifically for photorealistic human faces (trained on StyleGAN/StyleGAN2). Each prediction is accompanied by a visual heatmap generated by a dynamic Grad-CAM explainability module, explicitly highlighting the specific regions, textures, or artifacts that influenced the model's decision.

System Modules

1. Inference Engine Module: Dynamically routes the input image to the selected model (MobileNetV2 or EfficientNetV2B0), handles input dimensions, and handles necessary domain-specific preprocessing (e.g., MobileNetV2's [-1, 1] scaling).

2. Explainabili
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feature extract

Overall Workflow Diagram

Overall Workflow

The user uploads an image via the Streamlit interface and selects the appropriate model domain (general objects vs. faces). The frontend feeds the input to the Inference Engine, which scales and normalizes the image array according to the selected model's specifications. The image is passed through the model to compute the binary class probability (Real vs. AI Generated). Concurrently, the Explainability Module extracts feature maps from the final convolutional layer dynamically to produce a superimposed Grad-CAM overlay. Finally, the class prediction, confidence score, and explainability overlay are presented to the user.

CHAPTER 3: LITERATURE SURVEY

Research Papers Reviewed

1. *Deep Residual Learning for Image Recognition* (He et al.) - Provided the foundation for residual connections in deep convolutional networks.

2. *MobileNe
Explored for it

Existing Solutions

Current solutions often utilize massive Vision Transformers or frequency-domain analyzers. While highly accurate, they remain computationally expensive and lack granular, user-facing visual explainability.

Comparative Analysis

Compared to heavy Vision Transformers, the MobileNetV2 architecture requires significantly fewer parameters (approximately 3.4 million), making inference rapid on standard hardware. The explicit integration of Grad-CAM provides a layer of transparency frequently absent in commercial APIs.

Research Gap

A noticeable gap exists regarding lightweight, web-deployable synthetic image detectors that prioritize both inference speed and explicit visual explainability. The proposed system directly addresses this gap.

CHAPTER 4: TECHNOLOGIES USED

Programming Languages

- Python 3.9+

- CSS3

Frameworks and Libraries

- TensorFlow \& Keras:*- Utilized for model architecture construction, training, inference, and gradient calculations.

- Streamlit:*- I

Development Tools

- Git \& GitHub

- Visual Studic

CHAPTER 5: ARCHITECTURE AND DESIGN

System Architecture Diagram

Workflow Explanation

The system architecture decouples the frontend user interface from the backend inference and explainability engines. Streamlit manages the session state and input routing. The backend implements a thread-safe singleton design pattern for model instantiation, ensuring that once a model file is read into memory, it remains resident to reduce inference latency. The explainability module dynamically traverses the network layers to locate the Global Average Pooling layer, bypassing hardcoded layer references and rendering the Grad-CAM computation completely model-agnostic.

CHAPTER 6: IMPLEMENTATION

Key Features Implemented

1. Dynamic Model Loading: A singleton pattern ensures the model is loaded sequentially only once, reducing memory overhead and prediction latency.

2. Grad-CAM
to overlay jet-c

CHAPTER 7: RESULTS AND DISCUSSION

Output Analysis

The models compute binary classification labels indicating image authenticity (REAL or AI Generated). These labels are accompanied by a statistical confidence percentage derived directly from the model's final sigmoid activation output value.

Performance Evaluation

Both models were evaluated on their respective partition validation/testing sets. The performance parameters are detailed below:

- CIFAKE Model (MobileNetV2):*- Evaluated on a 20,000-image balanced test set.

- Validation A

- StyleGAN Model (EfficientNetV2B0):*- Evaluated on a balanced test partition from the 140,000 face dataset.

- Validation A

Table 7.1: CIFAKE Model Performance Metrics (Accuracy: 93.32%, Loss: 0.1690)

Table 7.2: StyleGAN Model Performance Metrics (Accuracy: 98.18%, Loss: 0.0498)

Testing Results

The application was subjected to functional testing. The CIFAKE model successfully identified standard objects, while the StyleGAN model successfully identified photorealistic facial deepfakes. During out-of-distribution testing (e.g., passing a human face to the CIFAKE model, or abstract objects to the StyleGAN model), it was observed that classification accuracy degrades, highlighting the importance of the dual-model selection feature implemented in the proposed system.

CHAPTER 8: CHALLENGES AND LEARNING OUTCOMES

Challenges Faced

1. Nested Model Gradients: The strict graph partitioning in Keras 3 complicated the extraction of the nested base model outputs, initially resulting in disconnected graph errors during Grad-CAM computation.

2. Out-of-Dis
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Solutions Implemented

1. Architectural Hooking: The Grad-CAM logic was refactored to interface with the input tensor of the subsequent layer (global average pooling), effectively extracting the required feature map while maintaining computational graph integrity.

2. Operational
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Technical Skills Learned

The project facilitated the acquisition of skills in advanced TensorFlow graph manipulation, GradientTape operations, Streamlit state management, and optimized inference pipeline design.

CHAPTER 9: FUTURE ENHANCEMENTS

Proposed Improvements

- Ensemble Modeling:*- Integrating the current architecture with an EfficientNet variant is proposed to improve generalization on out-of-distribution media.

- Dataset Exp
human faces a
everyday phot

Additional Features

- Batch Processing:*- Implementing parallel processing to allow users to verify bulk image datasets simultaneously.

Scalability Opportunities

The core inference logic can be abstracted into a REST API endpoint, permitting the model to serve mobile applications at scale via cloud containerization.

CHAPTER 10: CONCLUSION

Summary of Work Completed

The project successfully transitioned a trained MobileNetV2 architecture into a production-ready web application. It features a robust prediction pipeline, an integrated Explainable Artificial Intelligence module using Grad-CAM, and a formal user interface.

Achievements

The system maintained a strict 93.32% validation accuracy on the CIFAKE dataset while successfully mitigating complex nested-model gradient extraction errors. It delivers a visually striking and interactive application.

Project Impact

The developed application provides an educational and effective tool for identifying synthetic media. By prioritizing explainability, the project demystifies opaque artificial intelligence decisions and fosters critical analytical evaluation of digital authenticity.

Final Remark

As generative artificial intelligence capabilities advance, verification tools serve as a necessary mechanism in preserving digital authenticity. The architectural foundations established in this project are highly adaptable for future iterations against emerging synthetic media threats.